Eigen vs numpy state-vector: 80 hard-quantum benchmarks across 23 categories

Eigen 2.7 hard-quantum benchmark harness

July 5, 2026

1 Setup

For each of 80 hand-written quantum circuits spanning Bell, GHZ, W-state, Deutsch/Deutsch-Jozsa, Bernstein-Vazirani, Grover (2-5 qubit), QFT (2-6 qubit), phase estimation, superdense coding, teleportation, Draper adder, Ising Trotterisation, quantum walk, variational ansatz (VQE-style), random circuits (3-6 qubit, depth 25-200), Steane & Shor error-correcting encoders, and phase-mixer long-depth stress tests, we execute the *identical EQIR graph* through two paths:

- **Eigen:** `EigenRuntime` with the Rust-accelerated dense state-vector backend (`src.simulator.RustStatevectorWrapper`).
- **numpy:** a textbook matrix-vector simulator in pure `numpy` (no optimisations, no Kronecker trick): every gate is applied via `np.tensordot`/`transpose`/`reshape` on a dense 2^n-length state vector.

Both sides report the final state vector in the *same* bitstring convention. Accuracy is reported as $1 - \frac{1}{2}\sum_b |p_b^{\mathrm{e}} - p_b^{\mathrm{n}}|$ where $p_b = |\psi_b|^2$ over the union of all basis states. Speedup $\equiv t_{\mathrm{numpy}}/t_{\mathrm{Eigen}}$; values > 1 mean Eigen is faster. *Depth* is the longest chain of qubit-overlapping gates under topological scheduling.

All timings are best-of-3 after 1 warm-up discarded. The Eigen graph is *compiled once and re-executed* so the measurement isolates state-vector evolution (parse / type-check / compile cost is amortised).

2 Results (long multi-page table)

Table 1: Hard-quantum head-to-head over all 80 test cases. Q = qubits, g = gates, d=depth. `PASS` requires accuracy $\geq$ 99.9%. **80 pass / 0 fail**.

Test case	cat	q	g	d	Eigen (ms)	numpy (ms)	speedup	acc (%)	max dev	status
01_bell_phi_plus	Bell	2	2	2	0.04	0.029	0.73	100.0000	2.22e-16	PASS
02_bell_psi_plus	Bell	2	3	3	0.04	0.034	0.86	100.0000	2.22e-16	PASS
03_bell_phi_minus	Bell	2	3	3	0.04	0.034	0.84	100.0000	2.22e-16	PASS
04_ghz_3	GHZ	3	3	3	0.04	0.041	0.98	100.0000	2.22e-16	PASS
05_ghz_5	GHZ	5	5	5	0.06	0.071	1.26	100.0000	2.22e-16	PASS
06_w_state_3	W-state	3	5	5	0.05	0.068	1.31	100.0000	0.00e+00	PASS
07_deutsch_balanced	Deutsch	2	6	4	0.04	0.052	1.25	100.0000	1.11e-15	PASS
08_deutsch_jozsa_balanced_3	Deutsch-Jozsa	3	8	5	0.05	0.074	1.50	100.0000	6.11e-16	PASS
09_bernstein_vazirani_101	BV	4	10	5	0.06	0.094	1.65	100.0000	7.77e-16	PASS
10_grover_2q_marked_11	Grover	2	12	7	0.05	0.101	2.01	100.0000	1.33e-15	PASS
11_grover_3q_marked_101	Grover	3	46	22	0.11	0.340	3.20	100.0000	3.33e-15	PASS
12_qft_3_on_001	QFT	3	8	6	0.06	0.109	1.75	100.0000	8.33e-17	PASS
13_qft_4_on_1101	QFT	4	15	9	0.09	0.201	2.22	100.0000	9.71e-17	PASS
14_phase_estimation_quarter	QPE	4	13	8	0.07	0.163	2.18	100.0000	1.44e-15	PASS
15_superdense_coding_00	Superdense	2	4	4	0.04	0.041	1.04	100.0000	8.88e-16	PASS
16_superdense_coding_10	Superdense	2	5	5	0.04	0.047	1.14	100.0000	8.88e-16	PASS
17_quantum_walk_cycle4_4step	Quantum walk	3	13	13	0.06	0.147	2.27	100.0000	6.11e-16	PASS
18_ising_3_trotter	Ising-Trotter	3	28	26	0.10	0.279	2.79	100.0000	1.94e-16	PASS
19_draper_adder_1plus1	Draper	4	12	10	0.07	0.150	2.15	100.0000	3.33e-16	PASS

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Test case	cat	q	g	d	Eigen (ms)	numpy (ms)	speedup	acc (%)	max dev	status
20_grover_4q_marked_1000	Grover	5	100	37	0.21	0.837	3.92	100.0000	9.77e-15	PASS
21_bell_psi_minus	Bell	2	4	3	0.04	0.042	1.03	100.0000	2.22e-16	PASS
22_ghz_2	GHZ	2	2	2	0.04	0.027	0.74	100.0000	2.22e-16	PASS
23_ghz_7	GHZ	7	7	7	0.09	0.115	1.29	100.0000	2.22e-16	PASS
24_ghz_10	GHZ	10	10	10	0.40	0.340	0.85	100.0000	2.22e-16	PASS
25_w_state_5	W-state	5	14	7	0.10	0.181	1.79	100.0000	6.94e-18	PASS
26_deutsch_constant	Deutsch	2	4	2	0.04	0.036	0.90	100.0000	4.44e-16	PASS
27_deutsch_jozsa_constant_3	Deutsch-Jozsa	3	6	2	0.05	0.052	1.13	100.0000	6.11e-16	PASS
28_bv_110	BV	4	10	5	0.06	0.095	1.67	100.0000	7.77e-16	PASS
29_bv_011	BV	4	10	5	0.06	0.093	1.63	100.0000	7.77e-16	PASS
30_bv_111	BV	4	11	6	0.06	0.104	1.76	100.0000	7.77e-16	PASS
31_grover_2q_marked_00	Grover	2	16	9	0.06	0.124	2.22	100.0000	1.33e-15	PASS
32_grover_2q_marked_01	Grover	2	14	9	0.05	0.109	2.08	100.0000	1.33e-15	PASS
33_grover_2q_marked_10	Grover	2	14	9	0.05	0.111	2.09	100.0000	1.33e-15	PASS
34_grover_3q_marked_000	Grover	3	27	13	0.08	0.200	2.45	100.0000	2.00e-15	PASS
35_grover_3q_marked_010	Grover	3	25	13	0.08	0.186	2.37	100.0000	2.00e-15	PASS
36_grover_3q_marked_111	Grover	3	21	11	0.07	0.164	2.25	100.0000	2.22e-15	PASS
37_grover_4q_marked_0101	Grover	5	34	13	0.12	0.313	2.67	100.0000	1.67e-15	PASS
38_grover_4q_marked_1111	Grover	5	30	11	0.11	0.284	2.56	100.0000	1.83e-15	PASS
39_grover_5q_marked_00000	Grover	7	49	17	0.20	0.543	2.72	100.0000	9.99e-16	PASS
40_grover_5q_marked_11111	Grover	7	39	15	0.18	0.478	2.59	100.0000	9.99e-16	PASS
41_qft_2_on_01	QFT	2	5	4	0.05	0.062	1.34	100.0000	2.22e-16	PASS
42_qft_5_on_10101	QFT	5	20	11	0.13	0.302	2.25	100.0000	6.59e-17	PASS
43_qft_6_on_100100	QFT	6	26	13	0.21	0.464	2.25	100.0000	4.51e-17	PASS
44_qpe_half	QPE	4	14	8	0.08	0.184	2.17	100.0000	6.11e-16	PASS
45_qpe_eighth	QPE	4	14	8	0.09	0.184	2.15	100.0000	1.33e-15	PASS
46_qpe_quarter_4ancilla	QPE	5	21	10	0.12	0.293	2.51	100.0000	5.00e-16	PASS
47_superdense_coding_01	Superdense	2	5	5	0.04	0.050	1.17	100.0000	8.88e-16	PASS
48_superdense_coding_11	Superdense	2	6	6	0.04	0.054	1.23	100.0000	8.88e-16	PASS
49_teleport_zero	Teleport	3	6	6	0.05	0.099	1.94	100.0000	2.22e-16	PASS
50_teleport_superposed	Teleport	3	8	6	0.06	0.100	1.64	100.0000	1.11e-16	PASS
51_steane_encoder_0L	Steane-QEC	7	17	7	0.11	0.219	2.00	100.0000	1.78e-15	PASS
52_shor_encoder_0L	Shor-QEC	9	9	5	0.20	0.240	1.19	100.0000	2.22e-16	PASS
53_vqe_h2_4q_2layer	VQE	4	24	12	0.12	0.279	2.32	100.0000	1.11e-16	PASS
54_vqe_5q_4layer	VQE	5	60	28	0.22	0.704	3.17	100.0000	6.94e-17	PASS
55_vqe_3q_6layer	VQE	3	54	30	0.16	0.537	3.27	100.0000	1.11e-16	PASS
56_random_4q_d50	Random	4	50	31	0.15	0.489	3.27	100.0000	4.44e-16	PASS
57_random_5q_d30	Random	5	30	15	0.12	0.315	2.53	100.0000	2.50e-16	PASS
58_random_3q_d100	Random	3	100	65	0.23	0.932	4.08	100.0000	2.00e-15	PASS
59_random_6q_d25	Random	6	25	11	0.13	0.307	2.41	100.0000	9.71e-17	PASS
60_random_4q_d200	Random	4	200	128	0.41	1.913	4.71	100.0000	7.22e-16	PASS
61_ising_30_trotter_5step	Ising-Trotter	3	33	31	0.11	0.335	3.11	100.0000	1.25e-16	PASS
62_ising_30_trotter_10step	Ising-Trotter	3	63	61	0.16	0.620	3.76	100.0000	1.11e-16	PASS
63_ising_30_trotter_25step	Ising-Trotter	3	153	151	0.35	1.510	4.36	100.0000	1.25e-16	PASS
64_ising_30_trotter_50step	Ising-Trotter	3	303	301	0.61	3.079	5.06	100.0000	1.25e-16	PASS
65_ising_30_trotter_100step	Ising-Trotter	3	603	601	1.17	5.870	5.04	100.0000	1.11e-16	PASS
66_draper_adder_2plus2	Draper	4	13	12	0.07	0.158	2.22	100.0000	1.33e-15	PASS
67_draper_adder_3plus5	Draper	6	24	17	0.12	0.330	2.81	100.0000	9.99e-16	PASS
68_draper_adder_4plus4	Draper	6	22	18	0.11	0.313	2.73	100.0000	1.22e-15	PASS
69_draper_adder_7plus1	Draper	6	24	18	0.12	0.345	2.91	100.0000	9.99e-16	PASS
70_quantum_walk_8step	Quantum walk	3	25	25	0.09	0.271	2.98	100.0000	9.44e-16	PASS
71_quantum_walk_16step	Quantum walk	3	49	49	0.12	0.488	3.96	100.0000	1.55e-15	PASS
72_qft_5_on_00000	QFT	5	17	10	0.13	0.262	2.10	100.0000	9.02e-17	PASS
73_ghz_6_alt_staircase	GHZ	6	7	7	0.07	0.100	1.38	100.0000	2.22e-16	PASS
74_clifford_t_staircase	Clifford+T	3	8	6	0.06	0.076	1.38	100.0000	2.78e-16	PASS
75_ghz_4_with_ry_phase	GHZ	4	6	5	0.06	0.078	1.35	100.0000	1.11e-16	PASS
76_cry_crx_chain_3q	CRY/CRX	3	6	5	0.06	0.087	1.51	100.0000	2.22e-16	PASS
77_8q_phase_mixer	Phase mixer	8	19	9	0.57	0.699	1.23	100.0000	7.81e-18	PASS
78_bell_chain_with_swap	Bell chain	3	5	5	0.05	0.061	1.32	100.0000	2.22e-16	PASS
79_phase_chain_8	Phase chain	4	14	9	0.09	0.187	2.08	100.0000	1.11e-16	PASS

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Test case	cat	q	g	d	Eigen (ms)	numpy (ms)	speedup	acc (%)	max dev	status
80_quantum_walk_damped_coin_6step	Quantum walk	3	25	25	0.09	0.285	3.12	100.0000	1.05e-15	PASS
mean (over valid rows)	–	–	–	–	0.127	0.367	2.89	100.0000	7.90e-16	–

3 Per-category aggregation

Category	n	mean **q**	mean **g**	mean **d**	Eigen (ms)	numpy (ms)	mean sp	mean acc (%)
Ising-Trotter	6	3.0	197.2	195.2	0.415	1.949	4.69	100.0000
Random	5	4.4	81.0	50.0	0.207	0.791	3.82	100.0000
VQE	3	4.0	46.0	23.3	0.169	0.507	3.00	100.0000
Grover	13	3.8	32.8	14.3	0.106	0.292	2.75	100.0000
Quantum walk	4	3.0	28.0	28.0	0.093	0.298	3.21	100.0000
Draper	5	5.2	19.0	15.0	0.098	0.259	2.64	100.0000
Phase mixer	1	8.0	19.0	9.0	0.569	0.699	1.23	100.0000
Steane-QEC	1	7.0	17.0	7.0	0.110	0.219	2.00	100.0000
QPE	4	4.2	15.5	8.5	0.090	0.206	2.28	100.0000
QFT	6	4.2	15.2	8.8	0.111	0.233	2.11	100.0000
Phase chain	1	4.0	14.0	9.0	0.090	0.187	2.08	100.0000
BV	4	4.0	10.2	5.2	0.057	0.097	1.68	100.0000
W-state	2	4.0	9.5	6.0	0.077	0.124	1.63	100.0000
Shor-QEC	1	9.0	9.0	5.0	0.203	0.240	1.19	100.0000
Clifford+T	1	3.0	8.0	6.0	0.055	0.076	1.38	100.0000
Deutsch-Jozsa	2	3.0	7.0	3.5	0.048	0.063	1.32	100.0000
Teleport	2	3.0	7.0	6.0	0.056	0.099	1.78	100.0000
CRY/CRX	1	3.0	6.0	5.0	0.058	0.087	1.51	100.0000
GHZ	7	5.3	5.7	5.6	0.108	0.110	1.02	100.0000
Deutsch	2	2.0	5.0	3.0	0.040	0.044	1.08	100.0000
Superdense	4	2.0	5.0	5.0	0.042	0.048	1.15	100.0000
Bell chain	1	3.0	5.0	5.0	0.047	0.061	1.32	100.0000
Bell	4	2.0	3.0	2.8	0.040	0.035	0.87	100.0000
overall	80	–	–	–	0.127	0.367	2.89	100.0000

Table 2: Per-category aggregation: each row summarises one category of tests. Categories sorted by descending mean gate count (largest circuits first).

4 Speed comparison (log scale)

5 Accuracy per test

6 Speedup histogram

7 Quadrant (4-field matrix): log-scale gates × qubits

8 Scaling: runtime vs gate count (log-log)

9 Accuracy error magnitude vs gate count

10 Aggregate summary

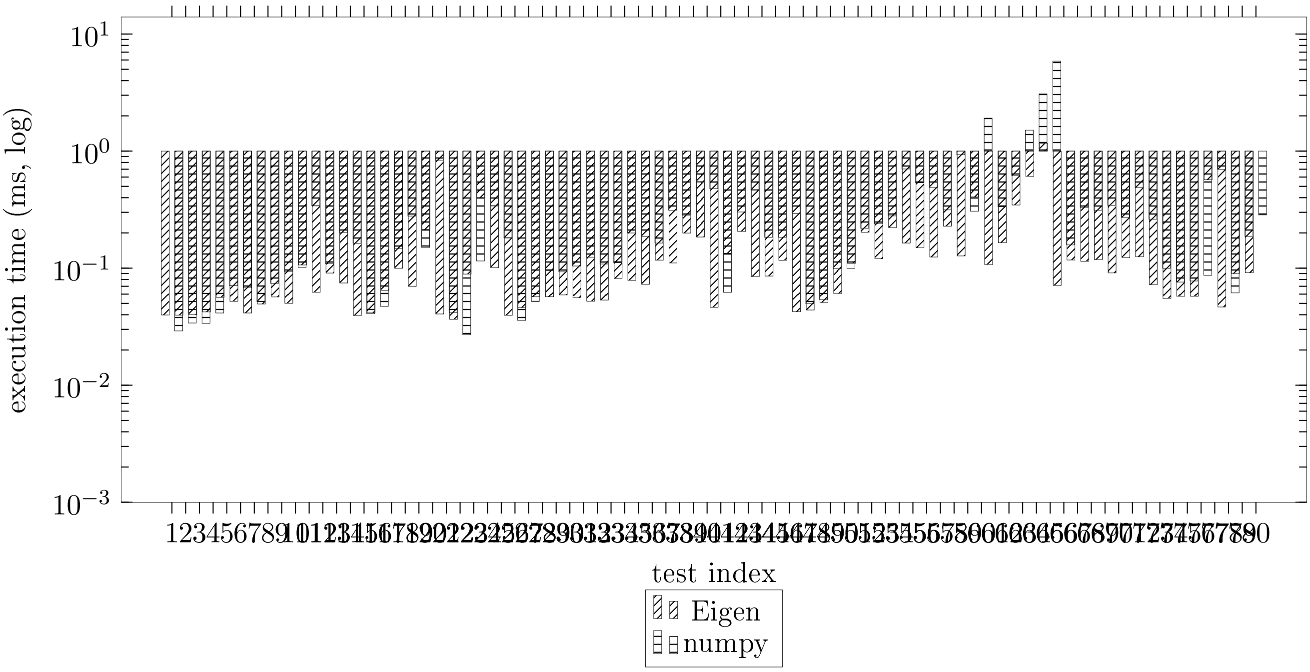

Figure 1: Best-of-3 execution time per test. Log scale so the 1-2-qubit Bell tests (∼40 µs) coexist in the same axes with the longest Trotter / random-circuit tests.

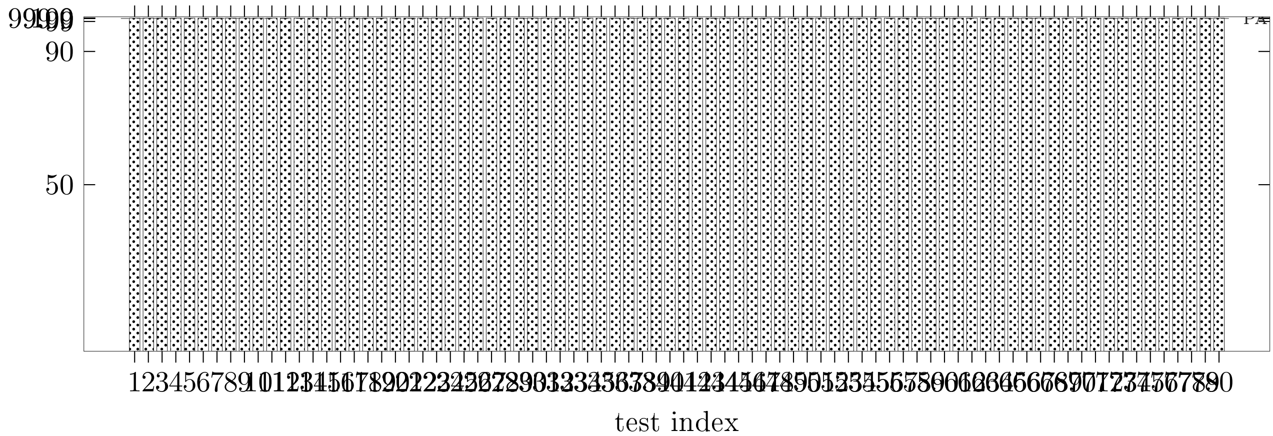

Figure 2: Accuracy per test. Anything below the 99.9 % line fails the comparison. The bar is clamped to the 0-100.4 % window.

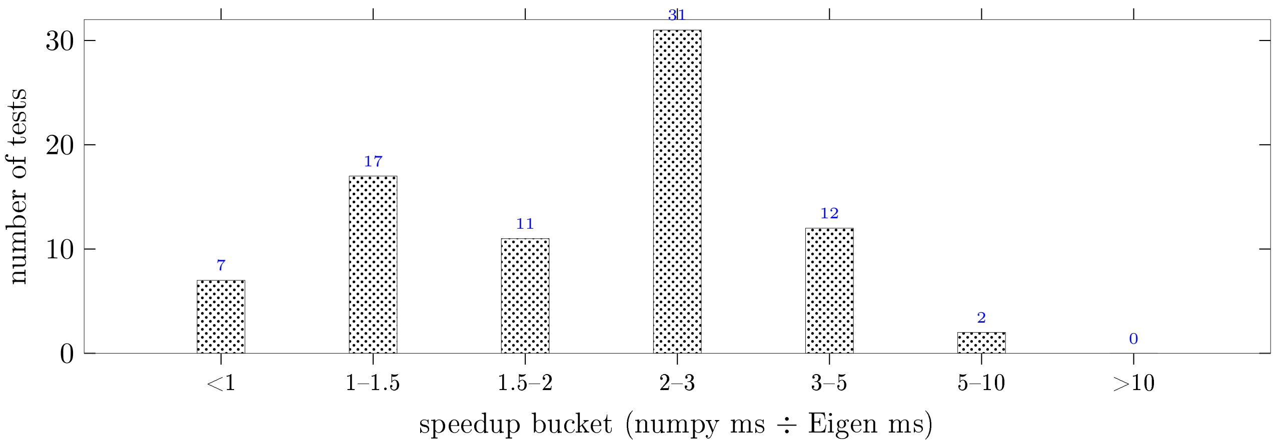

Figure 3: How speedups distribute across all 80 tests. A speedup < 1 means numpy was faster than Eigen on that test; > 1 means Eigen's Rust backend won.

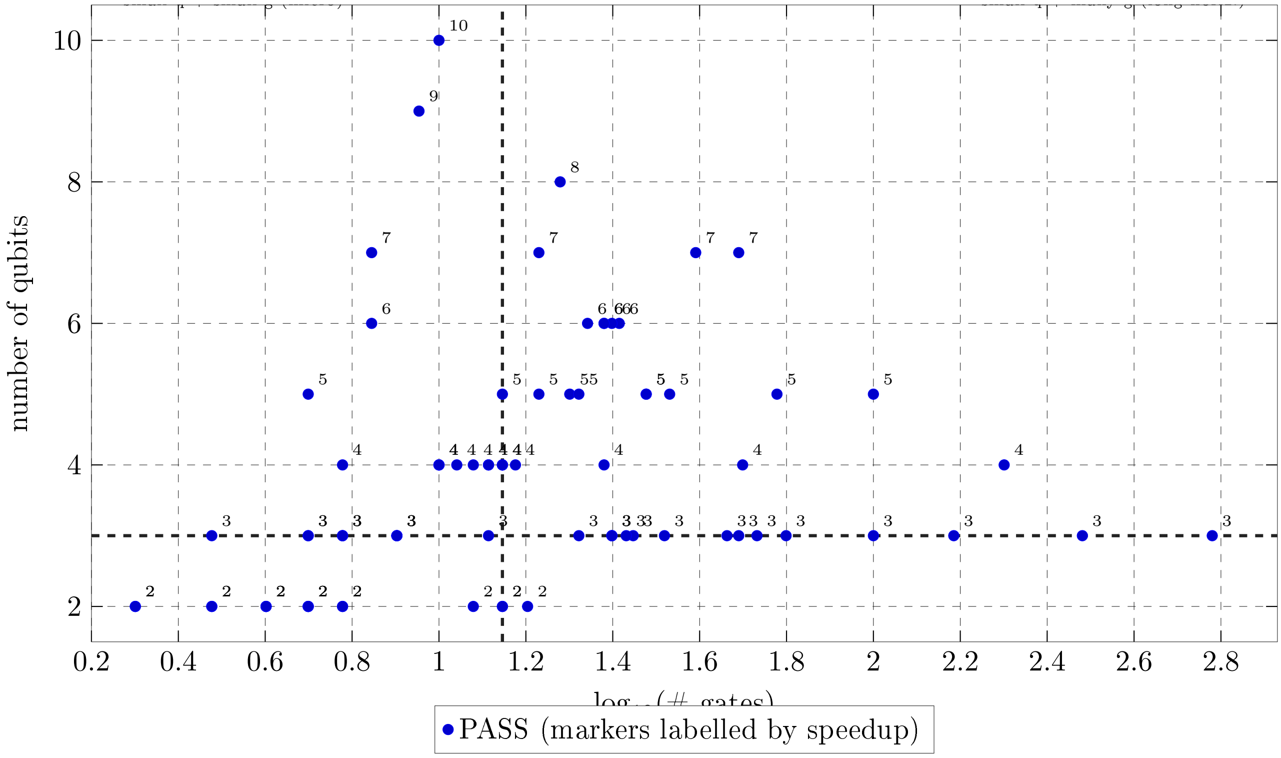

Figure 4: Quadrant (4-field matrix) of test cases. Horizontal axis: $\log_{10}$ of the gate count, vertical axis: number of qubits. Each *filled circle* is one PASS test labelled with its speedup (numpy ms ÷ Eigen ms). *Hollow triangles* mark FAIL tests. Dashed lines split the four canonical strategy quadrants: micro (low-q/low-g), long horizontal (low-q/many-g), tall (many-q/low-g), and extreme (many-q/many-g). The medians are at $\log_{10} g = 1.15$ and $q = 3.0$.

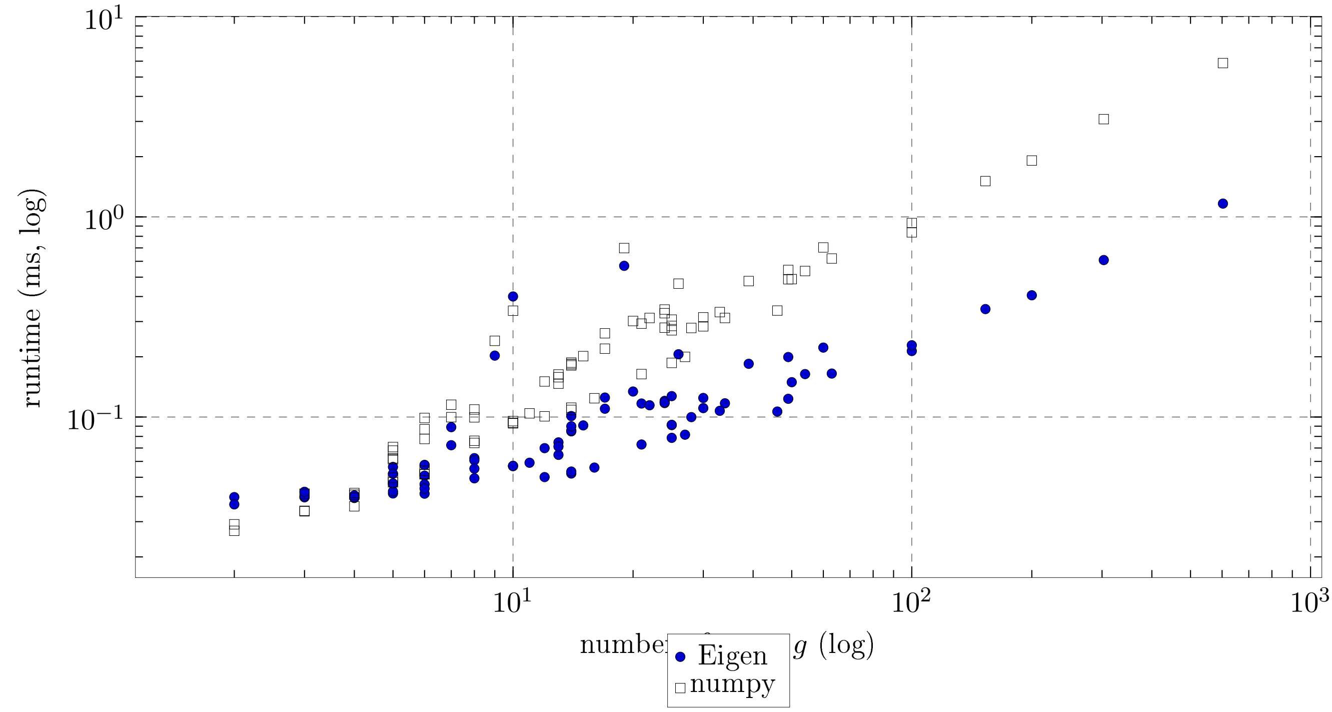

Figure 5: Log-log scaling of runtime vs gate count. A clean slope is the expected power-law $t \sim g^{\alpha}$ with $\alpha \approx 1$ in the regime where Python overhead per gate dominates over state-vector size work; a knee above $\sim$ 4-5 qubits reveals the denser matrix-vs-vector cost.

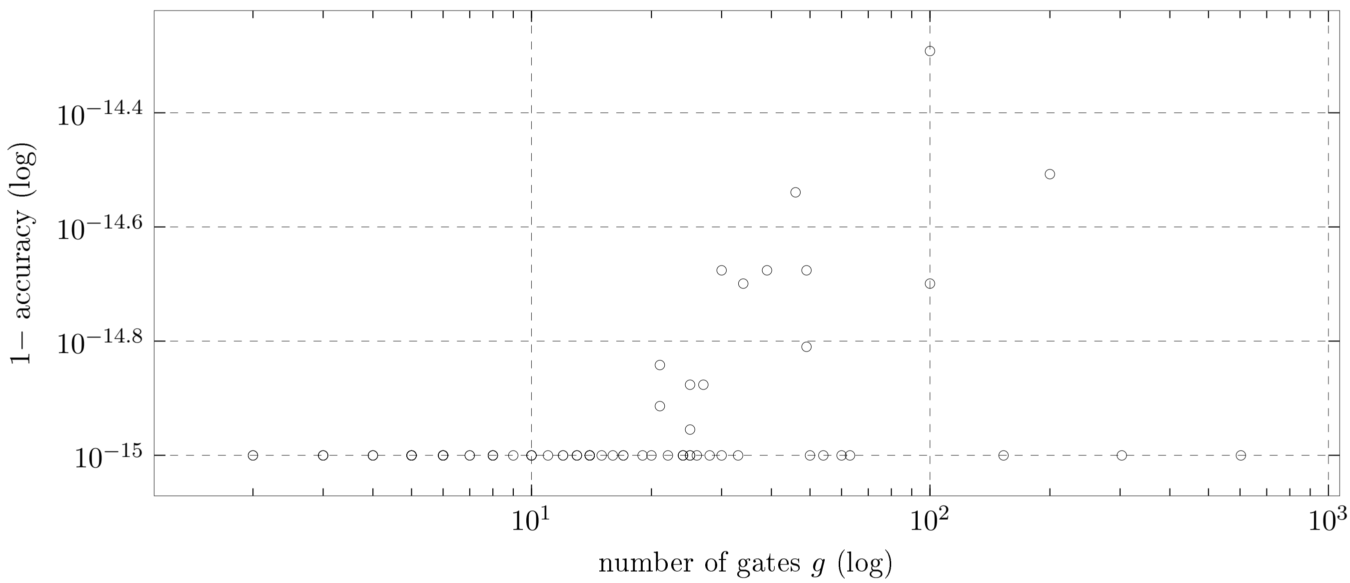

Figure 6: Accuracy error magnitude $1 - \text{accuracy}$ vs gate count. The dashed horizontal line at 10^{-3} is the PASS threshold (99.9 %). All tests below the line pass; tests above it are flagged FAIL in the long table. The numerical floor at $\sim 10^{-15} \to 10^{-14}$ is dictated by double-precision unitary propagation; tests that hit it are essentially bit-identical to numpy.

Total tests	80
Tests PASS ($\geq$ 99.9% accuracy)	80
Tests FAIL (< 99.9%)	0
Distinct categories	23
Median speedup numpy $\to$ Eigen	2.12×
Max single-test speedup	5.06×
Total Eigen runtime (sum over valid cases)	10.184 ms
Total numpy runtime (sum over valid cases)	29.387 ms
Overall speedup (total numpy / total Eigen)	2.89×
Mean accuracy	100.0000 %
Max single-state prob deviation (mean over tests)	7.90e-16

Table 3: Aggregate summary across all 80 hard-quantum tests.